

Offshore Fishing Hotspot — Species Targeting Reference

Scoring model v1.1 • species_config.json • All temperatures in °F • Depth in ft

Species	SST Target (Range)	Depth Ideal (Range) ft	CHL Range mg/m3	kd490 Max	Break Required	Min Zone Score
Yellowfin Tuna	73° (68–80°) <i>Warm side</i>	400–1,200 (300–3,000)	0.10 – 0.60	0.12	Yes	0.65
Mahi-Mahi	77° (72–86°) <i>Warm side</i>	200–1,200 (100–3,000)	0.05 – 0.40	0.1	No	0.58
Wahoo	75° (70–82°) <i>Warm side</i>	400–1,000 (200–1,500)	0.02 – 0.25	0.09	No	0.58
Bluefin Tuna	63° (55–70°) <i>Cold side</i>	400–1,500 (200–3,000)	0.15 – 1.50	0.2	Yes	0.6
King Mackerel	74° (68–80°) <i>Warm side</i>	60–180 (30–300)	0.20 – 2.00	0.3	No	0.6

Score Weights (must sum to 1.0 per species)

Species	SST	Break	Depth	CHL	Color
Yellowfin Tuna	0.35	0.30	0.20	0.10	0.05
Mahi-Mahi	0.30	0.30	0.15	0.10	0.15
Wahoo	0.30	0.20	0.25	0.10	0.15
Bluefin Tuna	0.35	0.30	0.18	0.12	0.05
King Mackerel	0.35	0.15	0.30	0.15	0.05

Scoring Logic

Factor	Formula / Behavior	None/Missing
SST	Gaussian peak at target; hard gate — point skipped if outside range	Point dropped
Break	Gradient magnitude: <0.4=none(0.15), 0.4-0.8=weak(0.45), 0.8-1.5=mod(0.75), >1.5=strong(1.0)	0.30 neutral
Depth	1.0 in ideal range; linear ramp to edges; 0.0 outside — hard gate if depth known	0.0 (not gated)
CHL	1.0 in range; partial credit just outside; 0.1 if zero	0.45 neutral
Color/kd490	1.0 if <=50% max; 0.75 if <=max; penalty beyond max	0.50 neutral
Break cap	If break_required=true and break_score<0.3: total capped at 0.50	—

Field Notes

Species	Notes
Yellowfin Tuna	Hard temperature breaks near canyon lips 300–1000 ft. Bait concentrations at chlorophyll edges. Blue water preferred but will follow breaks into transition zones. Peak season May–October Mid-Atlantic.
Mahi-Mahi	Warm side of color change lines. True blue water (kd490 < 0.10). Sargassum edges, floating debris. Pelagic — depth less critical than color break. SST min lowered to 72F to capture Hatteras/OBX transition zone fish.
Wahoo	Deep blue water near canyon edges and current boundaries. Less break-dependent than tuna. True offshore species — prefers very clear water.
Bluefin Tuna	Cold water side of breaks. Canyon edges, cold core eddies. Higher chlorophyll tolerance — follows bait into productive water. Peak: spring and fall migration.
King Mackerel	Nearshore and mid-range structure. Live bottom, ledges, color changes 30–300 ft. Not a strict blue-water species. Follows Spanish mackerel and baitfish inshore of the shelf break.

Data sources: VIIRS SST composite (daily, most-recent-wins) • ETOPO bathymetry • NASA MODIS Chlorophyll-a (8-day) • MODIS Kd490 sea color (8-day) • Gradient thresholds: weak 0.4°/°, moderate 0.8°/°, strong 1.5°/° lat/lon